

034IN -FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

FIRST PARTIAL EXAMINATION - A.Y. 2023/2024

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Problem 1 - From State-Space Equations to the Transfer Function

Tag List: [STANDARD ASSESSMENT] [PARTIAL CREDIT] [SYMBOLIC TOOLBOX]

Consider the LTI system

$$\begin{cases} \dot{x}(t) &= A x(t) + B u(t) \\ y(t) &= C x(t) + D u(t) \end{cases}$$

where the matrices are respectively:

$$A = \begin{bmatrix} -1 & -2 & -2 \\ -2 & -3 & +4 \\ +1 & -1 & -2 \end{bmatrix} \quad B = \begin{bmatrix} +6 \\ +8 \\ +1 \end{bmatrix} \quad C = [0 \quad +2 \quad +1] \quad D = [0]$$

Assignment:

- Determine the expression of the transfer function $G(s)$ of the given LTI system.
- Compute zeros, poles and the value of the system's static gain, exploiting the expression of $G(s)$.
- Determine the eigenvalues of the matrix A .
- Verify that the poles of $G(s)$ are a subset of the eigenvalues of the matrix A .
- **MANDATORY:** You must solve the assignment using suitable functions and commands from the *Symbolic Math Toolbox* **ONLY**.

Your solution has to:

- Create the **symbolic** matrices **A**, **B**, **C** and **D** according to the system description.
- Determine the eigenvalues of the symbolic matrix **A**, **sort** them in ascending order and store them in the **column symbolic array** named **eigA**.
- Obtain the **transfer function** of the system, by exploiting the expression $G(s) = C (sI - A)^{-1} B + D$.
- Assign the result in the MATLAB **symbolic function** (of course function of the symbolic variable **s** [the complex frequency of the Laplace transform]) named **G_s(s)**.
- Determine the **static gain**, the **zeros** and the **poles** of the transfer function $G(s)$. **You must use** the functions `poles( )`, `numden( )`, `solve(equation,var,"MaxDegree",3)` [refer to the MATLAB documentation] and exploit the properties of the symbolic functions.
- Assign the static gain to the **symbolic variable** **mu_G**.
- **Sort** zeros and poles in ascending order and then store the sorted zeros and poles in two **symbolic arrays**, named **zeros_SET** and **poles_SET**.
- **Verify** that the poles in **poles_SET** are also eigenvalues of A . In order to perform the check, use the function `ismember( )` and compare the contents of **eigA** (as first array) with **poles_SET** (as second array). If help is needed, refer to the [MATLAB documentation](#). Save the result of the comparison in the MATLAB array **LogicalFLAGS**.

Solution Proposal

```

syms s

% the matrix A
A = sym([-1 -2 -2; -2 -3 +4; +1 -1 -2]);

% the matrix B
B = sym([6; 8; 1]);

% the matrix C
C = sym([0 +2 +1]);

D = sym(0);

sI_A = s*eye(size(A))-A;
sI_Ainv = inv(sI_A);

eigA = sort(eig(A), 'ascend')

```

```

eigA =

$$\begin{pmatrix} -1-2i \\ -1+2i \\ -4 \end{pmatrix}$$


```

```

% the transfer function
G_s(s) = simplify(C * sI_Ainv * B +D)

```

```

G_s(s) =

$$\frac{17}{s+4}$$


```

```

% the poles of G(s)
poles_SET = sort(poles(G_s), 'ascend')

```

```

poles_SET = -4

```

```

% the zeros
[numG, ~] = numden(G_s)

```

```

numG(s) = 17

```

```

zeros_SET = sort(solve(numG, s, "MaxDegree",3), 'ascend')

```

```

zeros_SET =

Empty sym: 0-by-1

```

```

% the static gain - version 1
mu_G = G_s(0)

```

```

mu_G =

```

$$\frac{17}{4}$$

```
% the static gain - v2  
mu_GBIS = -C*inv(A)*B+D
```

```
mu_GBIS =
```

$$\frac{17}{4}$$

```
LogicalFLAGS = ismember(eigA, poles_SET)
```

```
LogicalFLAGS = 3x1 logical array  
0  
0  
1
```